Kinematics Modeling and Task-Space 3D Position Closed-Loop Control for a 6-DOF Serial Robot Manipulator

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***Abstract*—This paper presents an independent research on the kinematics modeling and task-space closed-loop control for a 6-DOF serial robot manipulator. The study adopts the Standard Denavit-Hartenberg (DH) convention to establish the kinematic model of a PUMA 560 manipulator. A geometrical Jacobian matrix is constructed. An adaptive Damped Least Squares (DLS) method is designed as a safety margin against kinematic singularities, but it is not triggered in this specific circular trajectory. To reduce the 3D position tracking error, a task-space closed-loop controller is designed with velocity feedforward and proportional position feedback. A comprehensive optimization is carried out, incorporating global velocity scaling and quintic time-scaling for smooth trajectory generation. Simulations are conducted in Python using the Robotics Toolbox. The final results demonstrate that the optimized controller achieves a maximum tracking error of 0.1393 mm, with the actual trajectory visually overlapping the circular reference trajectory.**

***Index Terms*—6-DOF serial manipulator, Denavit-Hartenberg, task-space control, damped least squares, trajectory tracking**

I. Introduction

Robot manipulator kinematics is the foundation of motion planning and control. For a 6-DOF serial manipulator, achieving precise task-space trajectory tracking is challenging due to nonlinearity, joint coupling, and kinematic singularities. Traditional joint-space control remains limited in directly addressing Cartesian requirements, whereas task-space control formulates the control objective directly in the operational space.

This paper studies a PUMA 560 industrial manipulator. The Standard DH convention is used to establish the forward kinematic model. An analytical geometric Jacobian is derived. A task-space closed-loop controller with velocity feedforward and proportional feedback is designed. An adaptive damping strategy is introduced as a safety margin to handle singularities. The performance is validated through 3D position tracking of a circular trajectory.

II. Kinematics Modeling

A. Standard DH Convention

The standard DH convention describes the transformation between consecutive link frames using four parameters: link twist α_{i-1}, link length a_{i-1}, joint offset d_i, and joint angle θ_i. The homogeneous transformation matrix from frame $i-1$ to frame i is:

$$ {}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_{i-1} & s\theta_i s\alpha_{i-1} & a_{i-1}c\theta_i \\ s\theta_i & c\theta_i c\alpha_{i-1} & -c\theta_i s\alpha_{i-1} & a_{i-1}s\theta_i \\ 0 & s\alpha_{i-1} & c\alpha_{i-1} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1} $$

The forward kinematic equation is obtained by successive multiplication:

$$ {}^0T_6 = {}^0T_1(\theta_1)\,{}^1T_2(\theta_2)\,{}^2T_3(\theta_3)\,{}^3T_4(\theta_4)\,{}^4T_5(\theta_5)\,{}^5T_6(\theta_6) \tag{2} $$

The PUMA 560 standard DH parameters used in this work are listed in Table I.

TABLE I
Standard DH Parameters of PUMA 560

Joint i	α_{i-1} (deg)	a_{i-1} (m)	d_i (m)	θ_i
1	90	0	0	θ_1
2	0	0.4318	0	θ_2
3	-90	0.0203	0.15005	θ_3
4	90	0	0.4318	θ_4
5	-90	0	0	θ_5
6	0	0	0	θ_6

B. Geometric Jacobian Matrix

The geometric Jacobian establishes the linear mapping between joint space velocity $\dot{q}$ and task-space Cartesian velocity $\dot{X}$:

$$ \Delta X = J(q)\Delta q \tag{3} $$

For each revolute joint, the i-th column of the full geometric Jacobian is:

$$ J_i = \begin{bmatrix} z_{i-1} \times (p_e - p_{i-1}) \\ z_{i-1} \end{bmatrix} \tag{4} $$

For the position Jacobian used in this work, only the linear part is retained:

$$ J_{v_i} = z_{i-1} \times (p_e - p_{i-1}) \in \mathbb{R}^3 \tag{5} $$

where z_{i-1} is the joint axis direction and p_{i-1} is the joint origin. Therefore, the position Jacobian is $J_v \in \mathbb{R}^{3\times 6}$.

III. Task-Space Controller Design

A. Quintic Time Scaling

To ensure zero initial and final velocities and accelerations, a quintic time scaling polynomial is adopted. The normalized time $s \in [0, 1]$ is defined as $s = t/T_{total}$. The smooth scaling function $h(s)$ and its derivative $\dot{h}(s)$ are:

$$h(s) = 10s^3 - 15s^4 + 6s^5 \tag{6}$$

$$\dot{h}(s) = \frac{30s^2 - 60s^3 + 30s^4}{T_{total}} \tag{7}$$

The desired Cartesian position p_d and feedforward velocity $\dot{p}_d$ are parameterized as:

$$p_d = p_{center} + \begin{bmatrix} -r\cos(2\pi h) \\ r\sin(2\pi h) \\ 0 \end{bmatrix} \tag{8}$$

$$\dot{p}_d = \begin{bmatrix} 2\pi r\dot{h}\sin(2\pi h) \\ 2\pi r\dot{h}\cos(2\pi h) \\ 0 \end{bmatrix} \tag{9}$$

where $r = 0.03$ m is the circular radius.

B. Feedforward Plus Proportional Feedback Control

A feedforward plus proportional feedback control law is designed for 3D position tracking. The integral term is omitted to prevent integral windup caused by large initial errors. The differential term is omitted to avoid high-frequency oscillation caused by error differentiation. The Cartesian velocity command is:

$$\dot{x}_{cmd} = \dot{p}_d + K_p e \tag{10}$$

where $e = p_d - p_{cur}$ is the 3D position error vector, and $K_p = 5.0 \cdot I_3$ is the diagonal proportional gain matrix.

C. Adaptive Damped Least Squares

To avoid excessive joint velocities near singularities, an adaptive damping factor λ is computed based on the minimum singular value σ_{min} of J:

$$\text{ratio} = \text{clip}\left(\frac{\sigma_{min}}{\sigma_{threshold}}, 0, 1\right) \tag{11}$$

$$\lambda = \lambda_{min} + (\lambda_{max} - \lambda_{min})(1 - \text{ratio}^2) \tag{12}$$

The joint velocity is resolved using the DLS formula:

$$\dot{q} = J^T(JJ^T + \lambda^2 I_3)^{-1}\dot{x}_{cmd} \tag{13}$$

where $\sigma_{threshold} = 0.03$, $\lambda_{min} = 10^{-4}$, and $\lambda_{max} = 0.02$. This mechanism acts as a safety margin; it is not triggered in this specific trajectory.

D. Global Speed Scaling

To preserve the direction of the end-effector motion, joint velocities are globally scaled instead of independently clipped:

$$\text{speed_scale} = \max\left(1.0, \frac{\max(|\dot{q}|)}{\dot{q}_{max}}\right) \tag{14}$$

$$\dot{q} \leftarrow \frac{\dot{q}}{\text{speed_scale}} \tag{15}$$

where $\dot{q}_{max} = 1.5$ rad/s. This ensures the ratio between joint velocities remains unchanged.

IV. Simulation Results

Simulations were conducted in Python using the Robotics Toolbox. The controller tracked a circular trajectory with a radius of 0.03 m. The simulation was conducted with a time step of $dt = 0.01$ s for a total of 5.0 s (500 steps). The initial joint configuration was set to the nominal pose (`puma.qn`). The joint angles were updated using Euler integration: $q_{k+1} = q_k + \dot{q}\Delta t$.

The performance metrics are summarized in Table II.

TABLE II
Tracking Error Metrics

Metric	Value
Maximum Position Error	0.1393 mm
RMS Position Error	0.0840 mm
Final Position Error	0.0137 mm
Speed Scaling Activation	0 / 500 steps
Minimum Jacobian Singular Value	0.239915

Fig. 1 shows the trajectory tracking results. The left panel shows the XY-plane trajectory, where the actual trajectory (blue) visually overlaps the desired circular trajectory (red dashed). The right panel shows the 3D position error norm over time in mm. The error initially rises due to the acceleration phase, reaches a peak in the middle section, and then gradually converges to a final value of 0.0137 mm.

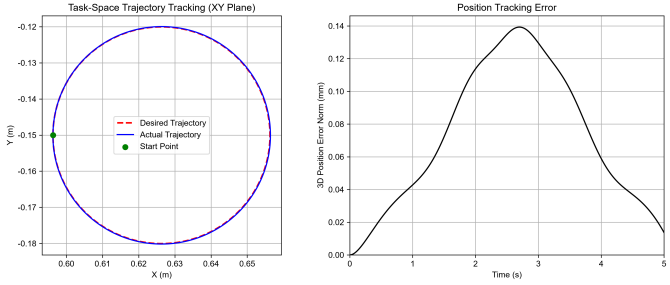


Fig. 1. Task-space trajectory tracking: XY-plane trajectory (left) and position error convergence (right).

During this specific circular trajectory, the minimum Jacobian singular value was 0.239915, which is well above the threshold of 0.03. As a result, the adaptive damping factor remained at its minimum value, and the global speed scaling was not triggered (0/500 steps). This indicates that the controller operated safely within this workspace.

V. Conclusion

This paper presented an independent study on kinematics modeling and task-space closed-loop control for a 6-DOF PUMA 560 manipulator. The Standard DH convention was used to establish the kinematic model. An adaptive DLS method was implemented as a safety margin against kinematic singularities. A feedforward plus proportional feedback controller with global speed scaling and quintic time scaling was designed for 3D position tracking. Simulation results demonstrated excellent tracking performance with a maximum position error of 0.1393 mm. Future work includes: (1) extending the controller to full 6-DOF pose control; (2) designing trajectories that intentionally approach singularities to fully validate the adaptive DLS; and (3) conducting ablation studies to compare pure P control, P+feedforward, and P+feedforward+DLS.

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